

Lesson 6 Introduction to PID Algorithm

1. PID Algorithm Description

PID algorithm is the most widely used autonomous controller which controls a process according to the ratio of errors, Proportional (P), Integral (I), and Derivative (D). With simple logic, it is easy to realize and can be widely applicable. And its control parameters are separate and it is easy to select the parameter.

$$u(t) = K_p \left(e(t) + \frac{1}{T_i} \int_0^t e(t) dt + T_d \frac{de(t)}{dt} \right)$$

Simplify:

$$u(t) = K_p e(t) + K_i \int_0^t e(t) dt + K_d \frac{de(t)}{dt}$$

And:

$u(t)$ ——the output of the PID controller;

K_p ——ratio factor;

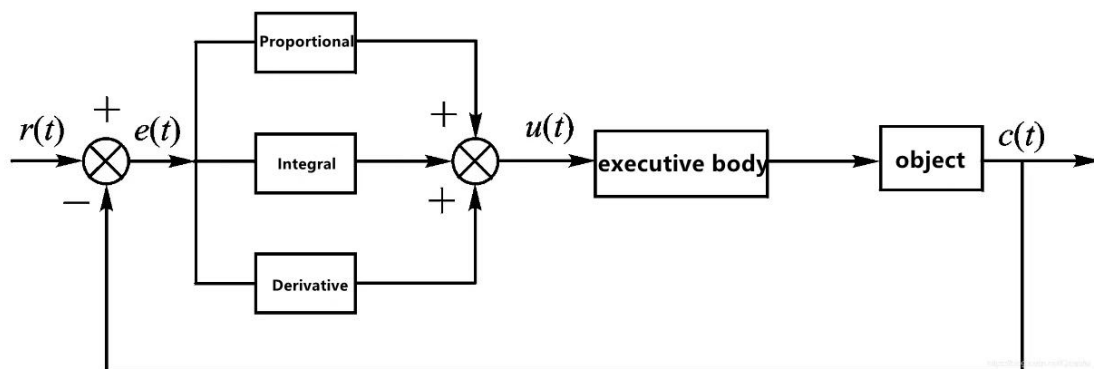
K_i ——integral time constant; $K_i = \frac{K_p}{T_i}$

K_d ——derivative time constant; $K_d = K_p \cdot T_d$

$e(t)$ ——the difference between the given value and the measured value (error)

2. PID Algorithm Logic

In a closed-loop control system, the output result of the object controlled will be returned to affect the output of the controller so as to generate one or multiple loop. The closed-loop system has positive and negative feedback. If the feedback signal is opposite to the given signal, it is negative feedback. If they are the same, it is positive feedback.



2.1 Proportional

Error is the difference between the given value (the set threshold $r(t)$) and the measured output ($c(t)$). After the proportional is introduced, the output of the system is proportional to the error, which can reduce the error quickly and stabilize the system. When the system tends to stabilize, steady-state error will still exist and cannot be eliminated. The control effect of proportional depends on the value of K_p . The smaller the proportional coefficient K_p , the weaker the control and the slower the system response. On the contrary, the larger the proportional coefficient K_p , the stronger the control, and the faster the system response. However, too large K_p will cause the system to generate large overshoot and oscillation. The advantage of proportional control lies in its fast response, but the disadvantage is that there is a steady-state error.

Take a bucket for example. There is a hole at the bottom of the bucket, but

you need to maintain the water in the bucket at 1m. And the current water level is 0.2m, and the water will leak 0.1m whenever you add the water. Here, proportional control can be adopted to control the water volume. The current water volume is 0.2m and the target is 1m. Set $K_p=0.4$. Then $u=0.4 \cdot e$, and $e = \text{last water volume} - \text{current water volume}$

The first error: $e(1)=1-0.2=0.8\text{m}$. Therefore, the volume of the added water is 0.32m ($K_p \cdot 0.8 = 0.4 \cdot 0.8 = 0.32\text{m}$), and the current water volume in the bucket is 0.42m ($0.2+0.32-0.1=0.42\text{m}$)

The second error: $e(2) = 1-(0.2+0.32) = 0.48\text{m}$. Therefore, the volume of the added water is 0.192m ($K_p \cdot 0.48=0.192\text{m}$), and the current water volume in the bucket is 0.552m ($0.52+0.192=0.552$)

NO.	Current difference	Output water volumn	Final output water volumn		KP
1			0.2		0.4
2	0.8	0.32	0.42		
3	0.58	0.232	0.552		
4	0.448	0.1792	0.6312		
5	0.3688	0.14752	0.67872		
6	0.32128	0.128512	0.707232		
7	0.292768	0.1171072	0.7243392		
8	0.2756608	0.11026432	0.73460352		
9	0.26539648	0.10615859	0.74076211		
10	0.259237888	0.10369516	0.74445727		
11	0.255542733	0.10221709	0.74667436		
12	0.25332564	0.10133026	0.74800462		
13	0.251995384	0.10079815	0.74880277		
14	0.25119723	0.10047889	0.74928166		
15	0.250718338	0.10028734	0.749569		
16	0.250431003	0.1001724	0.7497414		
17	0.250258602	0.10010344	0.74984484		
18	0.250155161	0.10006206	0.7499069		

Lastly, the water level stabilizes at 0.75m, and the error is 0.25m. When you add 0.1m water, the volume of the water added is equivalent to the volume

of the water leaking. And this kind of error is called steady-state error.

Steady-state error is defined as the difference between the output under the system tends to stabilize and the target output.

2.2 Integral

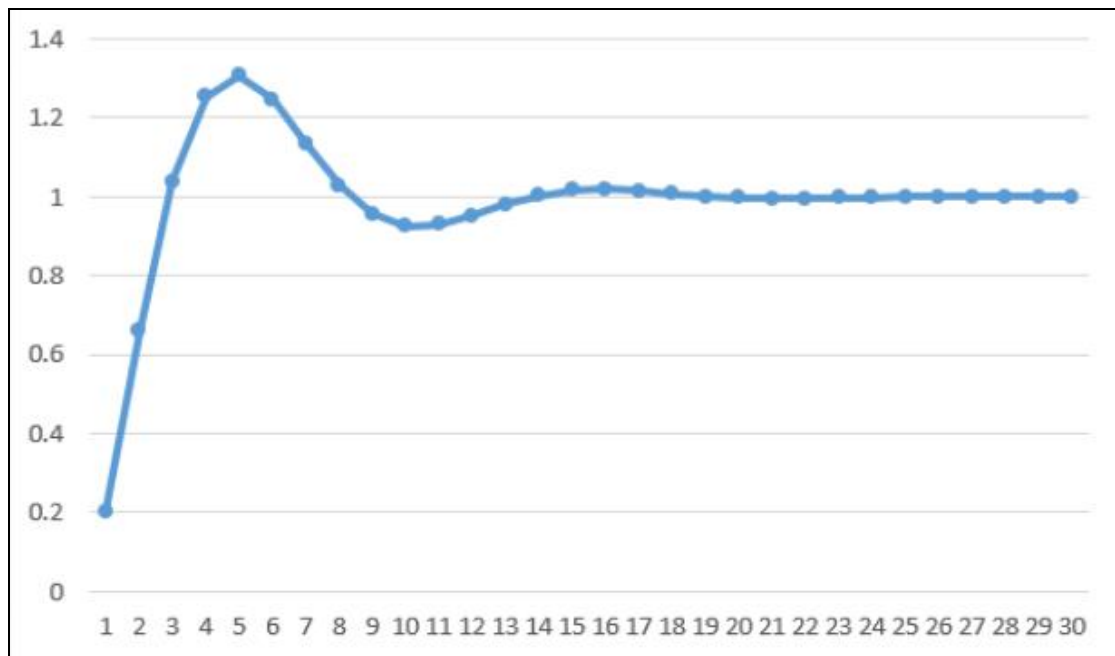
Integral aims at eliminating the steady-state error. The integral of the error over time is continuously accumulated to keep the output changing so as to reduce the error. If the integral time is sufficient, the steady-state error can be completely eliminated. The smaller T_i is, the stronger the function of integral is. But too strong integral function will make the system overshoot and even oscillate. The advantage of the integral is that the steady-state error can be completely eliminated, but the disadvantage is that the response speed is reduced and the overshoot will be too large.

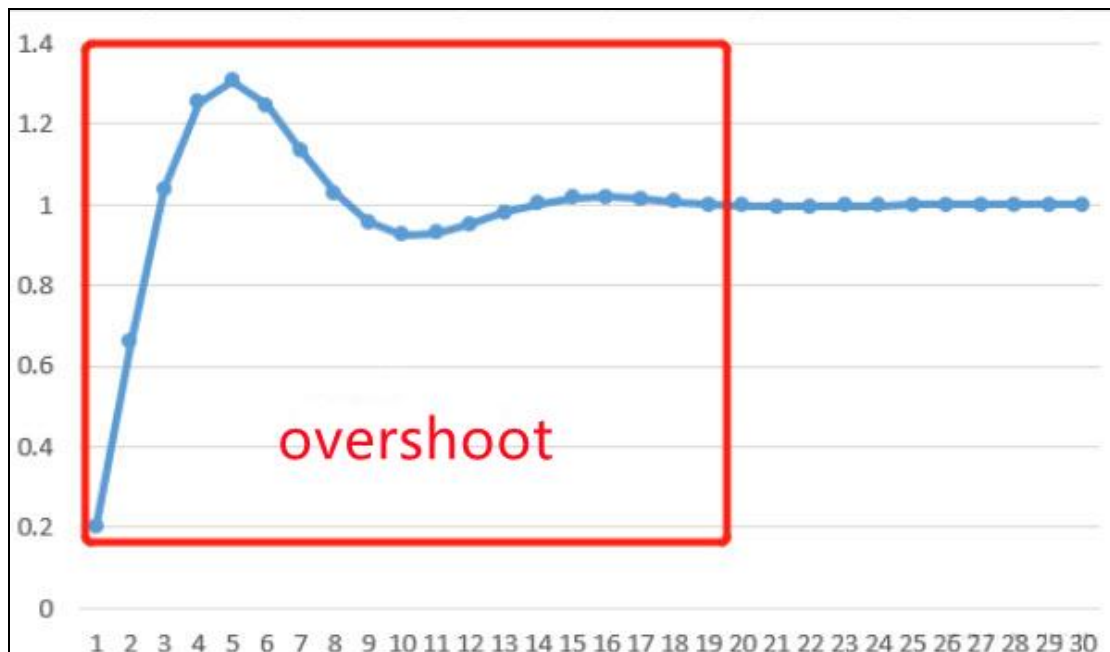
In the example given in “**2.1 Proportional**”, the steady-state error still exists after proportional control is adopted. To eliminate it completely, you can add integral control, and set the integral constant K_i as 0.3 ($K_i=0.3$).

The first error: $e(1)=1-0.2=0.8\text{m}$. Therefore, the volume of the added water is: $K_p \cdot 0.8 = 0.4 \cdot 0.8 = 0.32\text{m}$, $K_i \cdot e(1) = 0.3 \cdot 0.8 = 0.24$, and the final added water is: $0.32\text{m} + 0.256 = 0.56$. And the current water volume in the bucket is : $0.2 + 0.276 - 0.1 = 0.66$

The second error: $e(2)=1-0.66=0.34\text{m}$. Therefore, the volume of the added water is: $K_p \cdot 0.34 = 0.4 \cdot 0.34 = 0.136\text{m}$, $K_i \cdot (e(2)+e(1)) = 0.3 \cdot 0.342 = 0.342$, and the final added water is: $0.136\text{m} + 0.342 = 0.478$. And the current water volume in the bucket is : $0.66 + 0.478 - 0.1 = 1.038$

NO.	Current difference	KP output value	KI output value	PI output value	final water volumn	KP	KI
1					0.2		0.4
2	0.8	0.32	0.24	0.56	0.66		0.3
3	0.34	0.136	0.342	0.478	1.038		
4	-0.038	-0.0152	0.3306	0.3154	1.2534		
5	-0.2534	-0.10136	0.25458	0.15322	1.30662		
6	-0.30662	-0.122648	0.162594	0.039946	1.246566		
7	-0.246566	-0.0986264	0.0886242	-0.0100022	1.1365638		
8	-0.1365638	-0.0546255	0.04765506	-0.0069705	1.02959334		
9	-0.02959334	-0.0118373	0.03877706	0.02693972	0.95653306		
10	0.043466938	0.01738678	0.05181714	0.06920391	0.92573698		
11	0.074263023	0.02970521	0.07409605	0.10380126	0.92953823		
12	0.070461768	0.02818471	0.09523458	0.12341928	0.95295752		
13	0.047042484	0.01881699	0.10934732	0.12816432	0.98112183		
14	0.018878168	0.00755127	0.11501077	0.12256204	1.00368387		
15	-0.003683871	-0.0014735	0.11390561	0.11243206	1.01611593		
16	-0.016115934	-0.0064464	0.10907083	0.10262446	1.01874039		
17	-0.018740391	-0.0074962	0.10344871	0.09595256	1.01469295		
18	-0.014692948	-0.0058772	0.09904083	0.09316365	1.0078566		





After the Integral control is adopted, the steady-state error is completely eliminated, but overshoot occurs.

2.3 Derivative

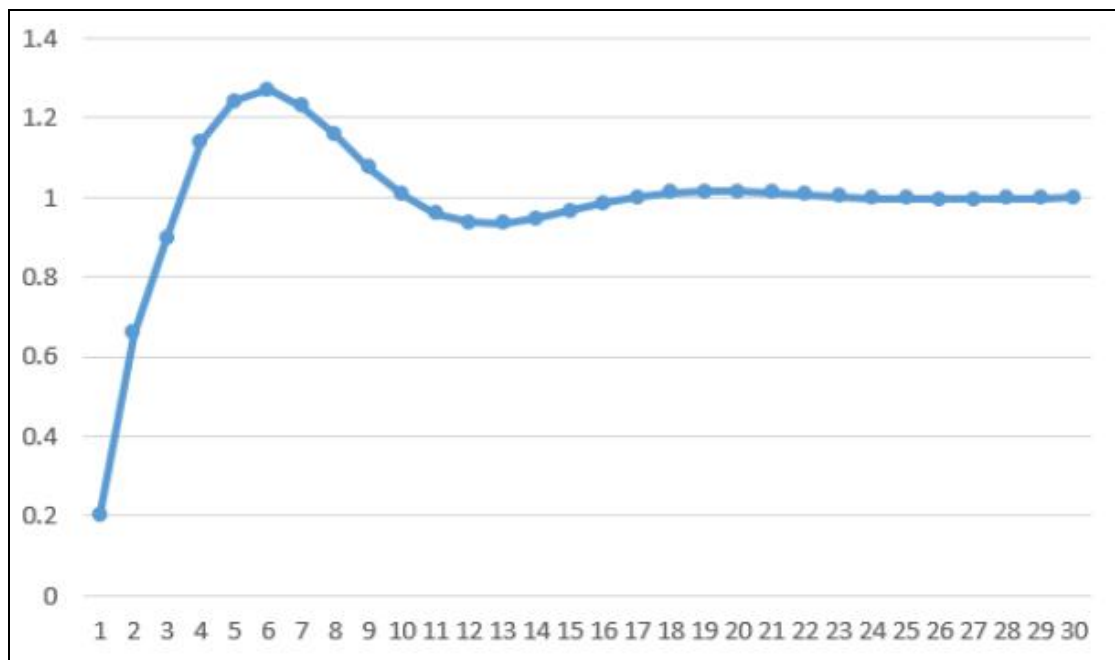
The slope error can be obtained by differentiating the error, and the slope reflects the change (rate of change) of the error signal, so the error can be adjusted in advance in derivative part. And it takes effect at the moment when the error appears or changes, so as to avoid excessive overshoot and reduce the dynamic error and response time. The larger the derivative coefficient T_d , the stronger the ability to suppress the error variation. The advantage of derivative is that it suppress overshoot and speed up the response. The disadvantage is that the anti-interference ability is poor. When the interference signal is strong, it is not recommended to add derivative.

In the above example, the integral effectively eliminates the steady-state error, but overshoot occurs. In order to reduce the overshoot, a derivative control is added. Set K_d as 0.3 ($K_d=0.3$).

The first error: $e(1)=1-0.2=0.8m$. Therefore, the volume of the added water is: $K_p \cdot 0.8 = 0.4 \cdot 0.8 = 0.32m$, $K_I \cdot e(1) = 0.3 \cdot 0.8 = 0.24$. And $K_D = 0$ (because the current water difference is 0.8). The final added water is: $0.32m + 0.256 = 0.56$. And the current water volume in the bucket is : $0.2 + 0.276 - 0.1 = 0.66$

The second error: $e(2)=1-0.66=0.34m$. Therefore, the volume of the added water is: $K_p \cdot 0.34 = 0.4 \cdot 0.34 = 0.136m$, $K_I \cdot (e(2)+e(1)) = 0.3 \cdot 0.342 = 0.342$. And $K_D \cdot (e(2) - e(1)) = 0.3 \cdot (0.342 - 0.8) = -0.138$. The final added water is: $0.136m + 0.342 - 0.138 = 0.34$. And the current water volume in the bucket is : $0.66 + 0.478 - 0.1 = 0.9$

NO.	current difference	KP output value	KI output value	KD output	PI output value	final water volumn	KP	KI	KD
1						0.2	0.4	0.3	0.3
2	0.8	0.32	0.24	0	0.56	0.66			
3	0.34	0.136	0.342	-0.138	0.34	0.9			
4	0.1	0.04	0.372	-0.072	0.34	1.14			
5	-0.14	-0.056	0.33	-0.072	0.202	1.242			
6	-0.242	-0.0968	0.2574	-0.0306	0.13	1.272			
7	-0.272	-0.1088	0.1758	-0.009	0.058	1.23			
8	-0.23	-0.092	0.1068	0.0126	0.0274	1.1574			
9	-0.1574	-0.06296	0.05958	0.02178	0.0184	1.0758			
10	-0.0758	-0.03032	0.03684	0.02448	0.031	1.0068			
11	-0.0068	-0.00272	0.0348	0.0207	0.05278	0.95958			
12	0.04042	0.016168	0.046926	0.014166	0.07726	0.93684			
13	0.06316	0.025264	0.065874	0.006822	0.09796	0.9348			
14	0.0652	0.02608	0.085434	0.000612	0.112126	0.946926			
15	0.053074	0.0212296	0.1013562	-0.0036378	0.118948	0.965874			
16	0.034126	0.0136504	0.111594	-0.0056844	0.11956	0.985434			
17	0.014566	0.0058264	0.1159638	-0.005868	0.1159222	1.0013562			
18	-0.0013562	-0.0005425	0.11555694	-0.0047767	0.1102378	1.011594			



Compared with proportional control, the overshoot is weakened.